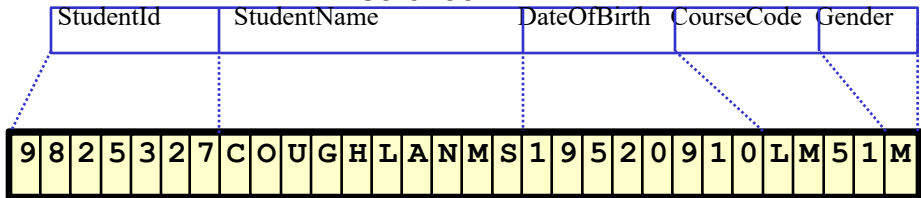


Declaring multiple record-type files

Click to view.

When viewing , left click (or press the Page Up key) to see the next frame and right click (or press Page Down) to see the previous frame.

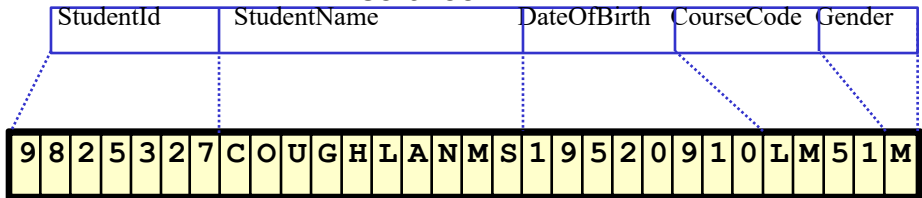
InsertRec



```
FD TransFile.  
01 InsertRec.  
    02 StudentId          PIC 9(7).  
    02 StudentName.  
        03 Surname        PIC X(8).  
        03 Initials       PIC XX.  
    02 DateOfBirth.  
        03 YOBirth        PIC 9(4).  
        03 MOBirth        PIC 99.  
        03 DOBirth        PIC 99.  
    02 CourseCode         PIC X(4).  
    02 Gender             PIC X.
```

When a file contains multiple record types, a record declaration (starting with a 01 level number) must be created for each type of record.

InsertRec



StudentId

DeleteRec
c

Even though there are different types of record in the file, and separate record declarations have been created for each record type, only a single 'record buffer' is created for the file.

```
FD TransFile.
01 InsertRec.
   02 StudentId          PIC 9(7) .
   02 StudentName.
      03 Surname         PIC X(8) .
      03 Initials        PIC XX .
   02 DateOfBirth.
      03 YOBirth         PIC 9(4) .
      03 MOBirth         PIC 99 .
      03 DOBirth         PIC 99 .
   02 CourseCode        PIC X(4) .
   02 Gender            PIC X .

01 DeleteRec.
   02 StudentId          PIC 9(7) .
```

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
-----------	-------------	-------------	------------	--------

9	8	2	5	3	2	7	C	O	U	G	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

StudentId

DeleteRec
c

StudentId

NewCourseCode

UpdateRec
c

All the record declarations map to this area of storage, which is the size of the largest record, and all the identifiers are current/active at the same time.

```
FD TransFile.
01 InsertRec.
  02 StudentId          PIC 9(7).
  02 StudentName.
    03 Surname          PIC X(8).
    03 Initials         PIC XX.
  02 DateOfBirth.
    03 YOBirth          PIC 9(4).
    03 MOBirth          PIC 99.
    03 DOBirth          PIC 99.
  02 CourseCode         PIC X(4).
  02 Gender             PIC X.
```

```
01 DeleteRec.
  02 StudentId          PIC 9(7).
```

```
01 UpdateRec.
  02 StudentId          PIC 9(7).
  02 NewCourseCode      PIC X(4).
```

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
-----------	-------------	-------------	------------	--------

9	8	4	1	3	2	5	L	M	6	2	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

StudentId

DeleteRec
c

StudentId

NewCourseCode

UpdateRec
c

When a record smaller than the size of the largest record is read into the buffer, any data that is not explicitly overwritten is left intact.

```
FD TransFile.
01 InsertRec.
  02 StudentId          PIC 9(7).
  02 StudentName.
    03 Surname          PIC X(8).
    03 Initials         PIC XX.
  02 DateOfBirth.
    03 YOBirth          PIC 9(4).
    03 MOBirth          PIC 99.
    03 DOBirth          PIC 99.
  02 CourseCode         PIC X(4).
  02 Gender             PIC X.
```

```
01 DeleteRec.
  02 StudentId          PIC 9(7).
```

```
01 UpdateRec.
  02 StudentId          PIC 9(7).
  02 NewCourseCode      PIC X(4).
```

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
-----------	-------------	-------------	------------	--------

9	6	2	7	3	9	2	L	M	6	2	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

StudentId

DeleteRec
c

StudentId

NewCourseCode

UpdateRec
c

When a record smaller than the size of the largest record is read into the buffer, any data that is not explicitly overwritten is left intact.

```
FD TransFile.
01 InsertRec.
  02 StudentId          PIC 9(7).
  02 StudentName.
    03 Surname          PIC X(8).
    03 Initials         PIC XX.
  02 DateOfBirth.
    03 YOBirth          PIC 9(4).
    03 MOBirth          PIC 99.
    03 DOBirth          PIC 99.
  02 CourseCode         PIC X(4).
  02 Gender             PIC X.
```

```
01 DeleteRec.
  02 StudentId          PIC 9(7).
```

```
01 UpdateRec.
  02 StudentId          PIC 9(7).
  02 NewCourseCode      PIC X(4).
```